

MIT integration bee - 2015

Notación y Consideraciones Generales

A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

Ejercicio 1

$$\int (\cos^4 x - \sin^4 x) dx$$

— — — Demostración — — —

$$\begin{aligned} I &= - \{ (\cos^2(x) - \sin^2(x))(\cos^2(x) + \sin^2(x)) \} \\ &= - \{ \cos(2x) \} \\ &= \frac{\sin(2x)}{2} + C \end{aligned} \quad (1)$$

- (1): $\cos(2x) = \cos^2(x) - \sin^2(x)$

Ejercicio 2

$$\int \frac{x}{\sqrt{2+4x}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \frac{1}{16} \left\{ \frac{u-2}{u^{1/2}} \right\} \\ &= \frac{1}{16} \left(\frac{2}{3} u^{3/2} - 4u^{1/2} \right) + C \end{aligned}$$

- (1): $\begin{cases} 2+4x &= u \\ 4dx &= du \end{cases}$

Ejercicio 3

$$\int_0^8 \frac{\cos \sqrt{x}}{\sqrt{x}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= 2 \{ \cos(t) \}_0^{2\sqrt{2}} \quad (1) \\ &= 2 (\sin(t))_0^{2\sqrt{2}} \\ &= 2 \sin(2\sqrt{2}) \end{aligned}$$

$$\bullet (1): \begin{cases} t &= \sqrt{x} \\ dt &= \frac{1}{2\sqrt{x}} dx \end{cases}$$

Ejercicio 4

$$\int \sec x dx$$

— — — Demostración — — —

$$I = \ln |\sec(x) + \tan(x)| + C$$

- (1): $\sec(x) = \frac{\sec(x)(\sec(x)+\tan(x))}{\sec(x)+\tan(x)}$
- (2): $[\sec(x) + \tan(x)] = \sec(x)(\sec(x) + \tan(x))$

Ejercicio 5

$$\int_0^{\pi/2} \frac{e^{\sin x}}{\tan x \csc x} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ e^{\sin(x)} \cos(x) \right\}_0^{\pi/2} \\ &= \left(e^{\sin(x)} \right)_0^{\pi/2} \quad (1) \\ &= e - 1 \end{aligned}$$

- (1): $[e^{f(x)}] = e^{f(x)} f'(x)$

Ejercicio 6

$$\int_1^e x(\log x)^2 dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \frac{1}{2} \left((\ln(x)x)^2 \right)_1^e - \{ x \ln(x) \}_1^e \quad (1) \\ &= \frac{e^2}{2} - \{ x \ln(x) \}_1^e \\ &= \frac{e^2}{2} - \left(\frac{\ln(x)x^2}{2} - \frac{x^2}{4} \right)_1^e \quad (2) \\ &= \frac{e^2}{4} - \frac{1}{4} \end{aligned}$$

- (1): $\left| \begin{array}{c} D \\ \ln^2(x) \\ \frac{2 \ln(x)}{x} \end{array} \right| \left| \begin{array}{c} I \\ x \\ \frac{x^2}{2} \end{array} \right|$

- (2): $\left| \begin{array}{c} D \\ \ln(x) \\ \frac{1}{x} \end{array} \right| \left| \begin{array}{c} I \\ x \\ \frac{x^2}{2} \end{array} \right|$

Ejercicio 7

$$\int \frac{1}{5 + 4\sqrt{x} + x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{2t}{5 + 4t + t^2} \right\} \\ &= \left\{ \frac{2t + 4}{5 + 4t + t^2} - \frac{4}{5 + 4t + t^2} \right\} \\ &= \left\{ \frac{2t + 4}{5 + 4t + t^2} - \frac{4}{(t + 2)^2 + 1} \right\} \\ &= \ln |5 + 4t + t^2| - 4 \arctan(t + 2) + C \end{aligned} \quad (1)$$

- (1): $\begin{cases} x &= t^2 \\ dx &= 2t dt \end{cases}$

Ejercicio 8

$$\int (2015)^x dx$$

--- Demostración ---

$$I = \frac{2015^x}{\ln(2015)} + C \quad (1)$$

- (1): $[a^x] = a^x \ln(a)$

Ejercicio 9

$$\int_0^2 \frac{x}{(x - 3)(x + 5)^2} dx$$

--- Demostración ---

$$\begin{aligned} I &= \frac{5}{8} \left\{ \left(\frac{3/5}{x - 3} + \frac{1}{x + 5} \right) \frac{1}{x + 5} \right\}_0^2 \\ &= \frac{5}{8} \left\{ \frac{3}{40} \left(\frac{1}{x - 3} - \frac{1}{x + 5} \right) + \frac{1}{(x + 5)^2} \right\}_0^2 \\ &= \frac{5}{8} \left(\frac{3}{40} (\ln |x - 3| - \ln |x + 5|) - \frac{1}{x + 5} \right)_0^2 \\ &= \frac{3}{64} \ln(5/21) + \frac{1}{28} \end{aligned}$$

Nota:

1. Hacer fracciones parciales para el producto de dos factores es más sencillo que determinar los coeficientes de toda la fracción parcial.

Ejercicio 10

$$\int \frac{\log(1 + \log x)}{x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \{ \log(1 + t) \} \\ &= (1 + t) \log(1 + t) - (1 + t) + C \\ &= (1 + t) \log(1 + t) - (1 + t) + C \\ &= \log(1 + \ln(x)) + \ln(x)(\ln(1 + \ln(x)) - 1) + C \end{aligned} \quad (1)$$

$$\bullet (1): \begin{cases} t &= \ln(x) \\ dt &= \frac{1}{x} dx \end{cases}$$

Ejercicio 11

$$\int \sqrt{\csc x - \sin x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \sqrt{\frac{1 - \sin^2(x)}{\sin(x)}} \right\} \\ &= \pm \left\{ \frac{\cos(x)}{\sqrt{\sin(x)}} \right\} \\ &= \pm 2\sqrt{\sin(x)} + C \end{aligned} \quad (1)$$

$$\bullet (1): [\sqrt{f(x)}] = \frac{f'(x)}{2\sqrt{f(x)}}$$

Ejercicio 12

$$\int \frac{1}{\sqrt{x^2 + 25}} dx$$

--- Demostración ---

$$\begin{aligned} I &= \frac{1}{5} \{ \sec(\theta) \} \\ &= \frac{1}{5} \ln |\sec(\theta) + \tan(\theta)| + C \\ &= \frac{1}{5} \ln \left| \frac{\sqrt{x^2 + 25}}{5} + \frac{x}{5} \right| + C \end{aligned}$$

$$\bullet (1): \begin{cases} x &= 5 \tan(\theta) \\ dx &= 5 \sec^2(\theta) d\theta \end{cases}$$

Ejercicio 13

$$\int_2^e \frac{\log^2 x - 1}{x \log^2 x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{t^2 - 1}{t^2} \right\}_{\ln(2)}^1 \quad (1) \\ &= \left(t + \frac{1}{t} \right)_{\ln(2)}^1 \\ &= -\frac{(\ln(2) - 1)^2}{\ln(2)} \end{aligned}$$

$$\bullet (1): \begin{cases} t &= \ln(x) \\ dd &= \frac{1}{x} dx \end{cases}$$

Ejercicio 14

$$\int e^{3x} \arctan e^x dx$$

--- Demostración ---

$$I = \frac{\arctan(e^x)e^{3x}}{3} - \frac{1}{3} \left\{ \frac{e^{4x}}{1 + e^{2x}} \right\} \quad (1)$$

$$= \frac{\arctan(e^x)e^{3x}}{3} - \frac{1}{6} \left\{ \frac{t}{1 + t} \right\} \quad (2)$$

$$= \frac{\arctan(e^x)e^{3x}}{3} - \frac{1}{6} (t - \ln |1 + t|)$$

$$= \frac{\arctan(e^x)e^{3x}}{3} - \frac{1}{6} (e^{2x} - \ln |1 + e^{2x}|) + C$$

$$\bullet (1): \left| \begin{array}{c} D \\ \arctan(e^x) \\ \frac{e^x}{1+e^{2x}} \end{array} \right| \left| \begin{array}{c} I \\ e^{3x} \\ \frac{e^{3x}}{3} \end{array} \right|$$

$$\bullet (2): \begin{cases} t &= e^{2x} \\ dt &= 2e^{2x} dx \end{cases}$$

Ejercicio 15

$$\int_0^4 \frac{|x-1|}{|x-2| + |x-3|}$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{x-1}{2x-5} \right\}_0^1 - \left\{ \frac{x-1}{2x-5} \right\}_1^2 + \{x-1\}_2^3 + \left\{ \frac{x-1}{2x-5} \right\}_3^4 \\ &= \frac{1}{2} \left\{ \frac{2x-2}{2x-5} \right\}_0^1 - \frac{1}{2} \left\{ \frac{2x-2}{2x-5} \right\}_1^2 + \{x-1\}_2^3 + \frac{1}{2} \left\{ \frac{2x-2}{2x-5} \right\}_3^4 \\ &= \frac{1}{2} \left\{ 1 + \frac{3}{2x-5} \right\}_0^1 - \frac{1}{2} \left\{ 1 + \frac{3}{2x-5} \right\}_1^2 + \{x-1\}_2^3 + \frac{1}{2} \left\{ 1 + \frac{3}{2x-5} \right\}_3^4 \\ &= \frac{1}{2} \left(x + 3 \frac{\ln |2x-5|}{2} \right)_0^1 - \frac{1}{2} \left(x + 3 \frac{\ln |2x-5|}{2} \right)_1^2 + \left(\frac{x^2}{2} - x \right)_2^3 + \frac{1}{2} \left(x + 3 \frac{\ln |2x-5|}{2} \right)_3^4 \\ &= \frac{9}{4} \ln(3) - \frac{3}{4} \ln(5) + 2 \end{aligned}$$

Ejercicio 16

$$\int_0^{2\pi} \frac{1}{\sin^4 x + \cos^4 x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{1}{(\sin^2(x) + \cos^2(x)) - 2 \sin^2(x) \cos^2(x)} \right\}_0^{2\pi} \\ &= \left\{ \frac{1}{1 - 2 \sin^2(x) \cos^2(x)} \right\}_0^{2\pi} \\ &= 2 \left\{ \frac{1}{2 - \sin^2(2x)} \right\}_0^{2\pi} \quad (1) \\ &= 2 \left\{ \frac{1}{2(\cos^2(2x) + \sin^2(2x)) - \sin^2(2x)} \right\}_0^{2\pi} \\ &= 2 \left\{ \frac{1}{\sin^2(2x) + 2 \cos^2(2x)} \right\}_0^{2\pi} \\ &= \left\{ \frac{\sec^2(2x)}{(\tan(2x)/\sqrt{2})^2 + 1} \right\}_0^{2\pi} \\ &= \sum_{n=0}^7 \left\{ \frac{\sec^2(2x)}{(\tan(2x)/\sqrt{2})^2 + 1} \right\}_{n\pi/4}^{(n+1)\pi/4} \\ &= \frac{1}{\sqrt{2}} \left(\left\{ \frac{1}{t^2 + 1} \right\}_0^{+\infty} + \left\{ \frac{1}{t^2 + 1} \right\}_{-\infty}^{+\infty} + \left\{ \frac{1}{t^2 + 1} \right\}_{-\infty}^{+\infty} + \left\{ \frac{1}{t^2 + 1} \right\}_{-\infty}^{+\infty} + \left\{ \frac{1}{t^2 + 1} \right\}_{-\infty}^0 \right) \quad (2) \\ &= \frac{1}{\sqrt{2}} \left((\arctan(t))_0^{+\infty} + 3 (\arctan(t))_{-\infty}^{+\infty} + (\arctan(t))_{-\infty}^0 \right) \\ &= 2\sqrt{2}\pi \end{aligned}$$

• (1): $\sin(2x) = 2 \sin(x) \cos(x)$

• (2): $\begin{cases} t = \frac{\tan(2x)}{\sqrt{2}} \\ dt = \sqrt{2} \sec^2(2x) dx \end{cases}$

Ejercicio 17

$$\int \frac{1 + e^x}{1 - e^x} dx$$

--- Demostración ---

$$\begin{aligned} I &= 2 \left\{ \frac{1 + e^{2t}}{1 - e^{2t}} \right\} \quad (1) \\ &= -2 \left\{ \frac{\cosh(t)}{\sinh(t)} \right\} \\ &= -2 \ln |\sinh(t)| + C \\ &= -2 \ln |\sinh(x/2)| + C \end{aligned}$$

• (1): $\begin{cases} x = 2t \\ dx = 2 dt \end{cases}$

Nota:

1. Propiedades básicas de las funciones trigonométricas hiperbólicas:

$$\left\{ \begin{array}{lcl} \cosh^2(x) - \sinh^2(x) & = & 1 \\ 2 \sinh^2(x) & = & \cosh(2x) - 1 \\ 2 \cosh^2(x) & = & \cosh(2x) + 1 \\ \sinh^2(x) + \cosh^2(x) & = & \cosh(2x) \\ 2 \sinh(x) \cosh(x) & = & \sinh(2x) \\ (\sinh(x) + \cosh(x))^2 & = & \cosh(2x) + \sinh(2x) \end{array} \right.$$

2. Derivadas de las funciones trigonométricas hiperbólicas:

$$\left\{ \begin{array}{lcl} [\sinh(x)] & = & \cosh(x) \\ [\cosh(x)] & = & \sinh(x) \\ [\tanh(x)] & = & \operatorname{sech}^2(x) \\ [\coth(x)] & = & -\operatorname{csch}^2(x) \\ [\operatorname{sech}(x)] & = & -\operatorname{sech}(x)\tanh(x) \\ [\operatorname{csch}(x)] & = & -\operatorname{csch}(x)\coth(x) \end{array} \right.$$

Ejercicio 18

$$\int \tan^4 x dx$$

--- Demostración ---

$$\begin{aligned} I &= \{ \tan^2(x) \sec^2(x) - \sec^2(x) + 1 \} \quad (1) \\ &= \frac{\tan^3(x)}{3} - \tan(x) + x + C \end{aligned}$$

$$\bullet (1): \tan^2(x) + 1 = \sec^2(x)$$

Ejercicio 19

$$\int \sin x \tan^2 x dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{\sin(x)(1 - \cos^2(x))}{\cos^2(x)} \right\} \\ &= \left\{ \frac{u^2 - 1}{u^2} \right\} \quad (1) \\ &= u + \frac{1}{u} + C \\ &= \cos(x) + \sec(x) + C \end{aligned}$$

$$\bullet (1): \begin{cases} u &= \cos(x) \\ du &= -\sin(x) dx \end{cases}$$

Ejercicio 20

$$\int \frac{x+1}{x^2+2x+3} dx$$

--- Demostración ---

$$\begin{aligned}
I &= \frac{1}{2} \left\{ \frac{2x+2}{x^2+2x+3} \right\} \\
&= \frac{1}{2} \ln |x^2+2x+3| + C
\end{aligned}$$

$$\bullet \text{ (1): } [\ln(f(x))] = \frac{f'(x)}{f(x)}$$